

When Passes Bend: Trajectory-Aware Pass Completion Modeling Through 3D Curvature and Defensive Reachability

Introduction: Expected pass completion (xC) is now central to pass evaluation and spatial-value modeling, estimating the likelihood a pass reaches its target without interception. Early models relied on raw completion rates, evolving to incorporate pass characteristics like location, distance, angle, and type ([Power, 2017](#)), physics-based ball and player trajectories ([Spearman, 2017](#)), and contextual factors such as pressure, body part, and possession state ([Gallagher, 2021](#)). While recent xPass models combine spatiotemporal tracking and physics-based trajectories to assess pass difficulty ([Anzer, 2022](#)), interception models still largely assume straight-line ball paths. This overlooks how 3D curvature and height, shaped by passer skill, can bend passes around or over defenders, giving identical endpoints different interception risks. We introduce a trajectory-aware framework quantifying how 3D curvature and height affect pass completion, positioning actual ball trajectory—not just endpoint geometry—as a distinct dimension of pass valuation.

Methods

Dataset: We analyzed optical ball tracking (timestamps; ball x, y, z) from the [public World Cup 2022 tracking and event data](#). A total of 40,440 passes with valid 3D trajectories were extracted along with their starting positions and outcomes. Each pass received a normalized 3D curvature score based on the actual ball 3D trajectory and its straight start-to-end chord. Based on that, a look up table is prepared with the probabilities of the passes curvature and where they originate on the pitch, along with the pass outcome given its curvature.

Interception Model: To estimate interception probability for an arbitrary pass, we model a 3D elliptical volume with the start and end points as its main axis vertices. Within this volume, 100 candidate 3D arcs connecting the two points are sampled, each subdivided into line segments based on arc length. The standard straight-line interception model ([Spearman, 2017](#)) is applied to each segment, adjusted for defender height, and aggregated across the arc. The final interception probability is a weighted average across all sampled arcs, weighted by the likelihood of that arc being followed given its curvature and pitch origin.

Spatial Models Evaluation: To evaluate our proposed model, we integrated its calculation in the Pitch Control (PC) ([Spearman, 2016](#)), Off-Ball Scoring Opportunity (OBSO) ([Rios-Neto, 2020](#)) and Dangerous Accessible Space (DAS) ([Bischofberger, 2026](#)) spatial models.

Results: Figure 1.a,b illustrates the prevalence and curvature diversity of curved passes across all passes and different pitch regions. Figure 1.c-e shows the interception map construction for an example pass. Figure 2 shows examples of PC, OBSO and DAS maps when using the conventional straight-line and proposed curved pass models.

Conclusion: Curved passes are common in soccer yet largely overlooked in pass outcome models. By incorporating curvature, ball height, and defensive reachability, our framework captures interception risks that straight-line models miss and shows how alternative trajectories can shift completion probability. These findings underscore the value of trajectory-aware modeling for more realistic pass valuation and spatial analysis in soccer.

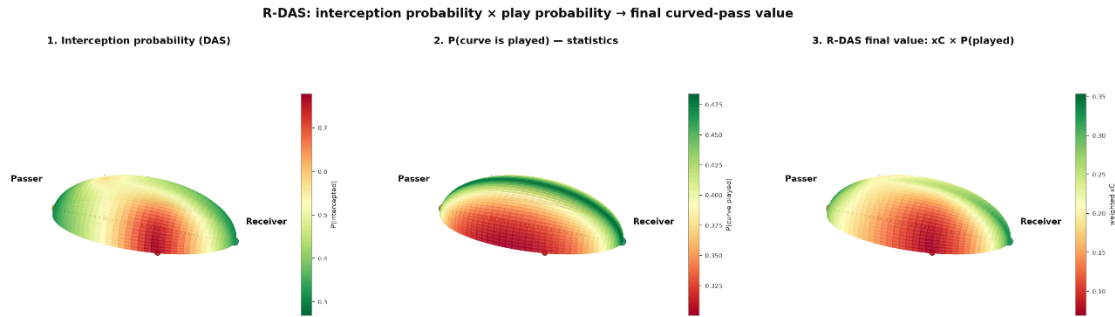
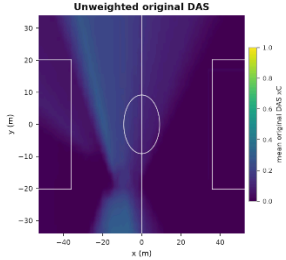
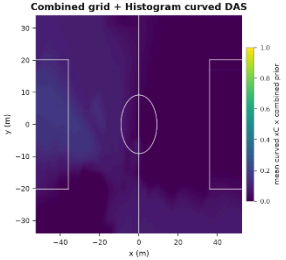
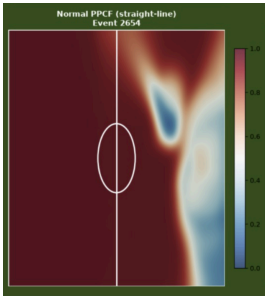
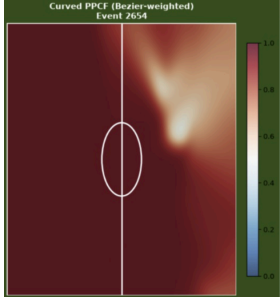
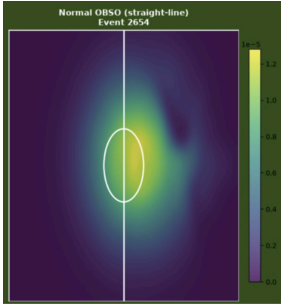
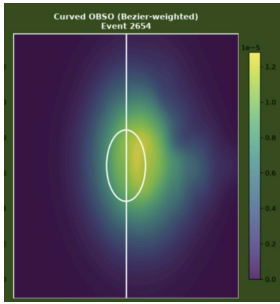
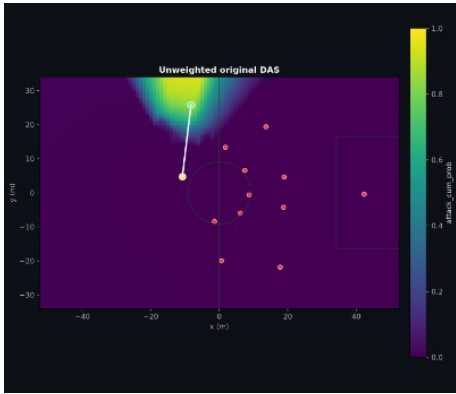
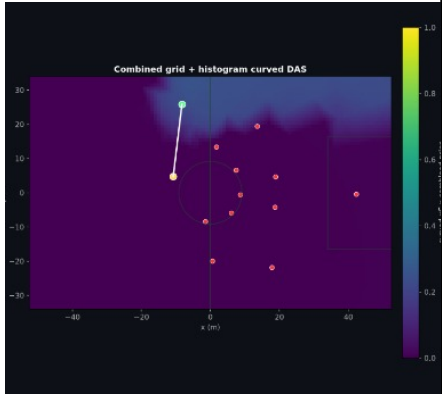


Figure 1. a) Histogram of all passes based on their curvature (y-axis is in log scale). Each bar is split based on successful and failed percentage of the bar total number of passes, b) Pitch map for the percentage of curved passes at each rectangle of the pitch, c-e shows the interception map construction for an example pass.

Comment (to be removed):

Fig1: a) Histogram of all pass based on their curvature and with a log scale for the y-axis. Color the straight passes bar with a different color and write its % on top of it. Also, can we split each bar into two colors based on the successful and failed percentage of the bar total pass, b) Pitch map for the percentage of curved passes at each rectangle of the pitch.

		Conventional Maps "Straight-line model"	Proposed adjusted maps "Curved pass model"
Example1	Pitch Control		
	OBSO		

	DAS		
Example2	Pitch Control		
	OBSO		
	DAS		
Histograms	Pitch Control		
	OBSO		

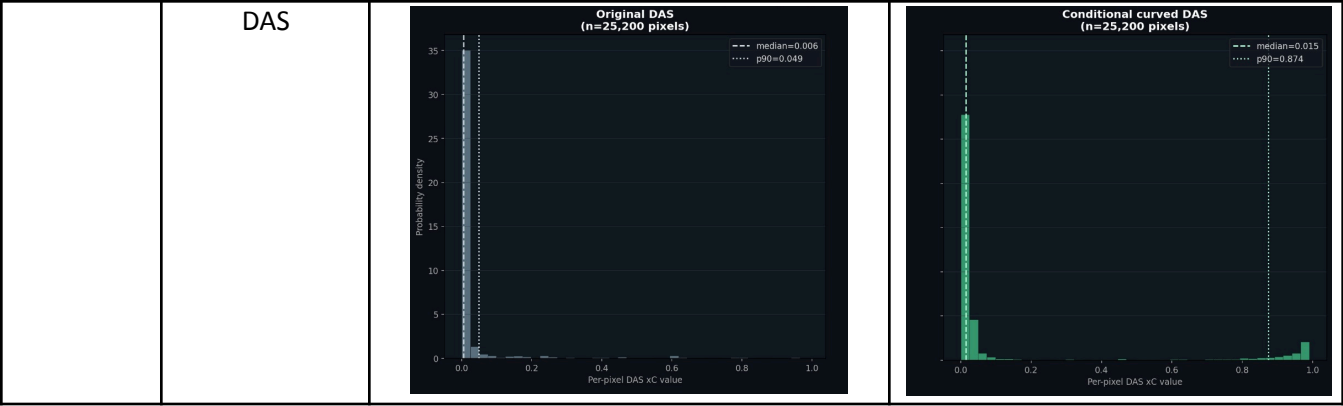


Figure 2. Model prediction error per player when testing on the players of team France (to the left), and when testing on the player of a different team (to the right). The prediction error changes significantly within the different players' (mostly based on the player tactical positions). Moreover, being trained on the seven games of France, the model exhibits lower error for the French team compared to the other team (Overall MAE 6.4m vs. 16.94m).